

PERFORMANCE EVALUATION OF NOVEL SECURE KEY MANAGEMENT SCHEME OVER BAN WIRELESS SENSOR NETWORKS

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ABSTRACT

Recent researches have witnessed that secret keying functionality is very important to protect wireless sensor networks from existing adversary. Based on inherited nature and limited resources capabilities of wireless sensor devices, providing a strong keying scheme is very difficult. Existing researches in literature within security domains for sensor network introduced several security solutions. This paper proposed performance evaluation of robust and secure keying functionality scheme, which provides robust and scalable post-deployment key generation. In this context, the proposed scheme helped the hierarchically clustered based topology platform to provide feasible and multi-level security tasks. This study considered three existing security techniques TinySec, MiniSec, and SHARP from packet overheads perspective and compared the time to transmit, average latency, and energy expenditure. Based on comparisons in analytical and experimental findings, the packet overhead, time to transmit, latency overhead, and energy overhead in NSKM module are less compared to other existing schemes in the literature. Based on the current study findings, this work is the first performance evaluation carried out for proposed secured Body Area Network (BAN) using wireless sensor devices. This study designed secure scheme that provides robust action to face sensor node compromising attacks. Besides, this study introduced lightweight, multi-level security services and scalable to network topology changes. Also this work outperformed other existing schemes in literature. Furthermore, the proposed platform was feasible to be implemented in dynamic and large size scale wireless sensor networks.

KEY WORDS: Security, Key creation, Hierarchically clustered networks, Super cluster head, Cluster head, key management, BAN, Packet overheads, Latency.

1. INTRODUCTION

Applications implementing upon WSNs are viable for a substantial attendance in the future. Security is essential in some monitoring scenarios, disaster relief management, health care field and monitoring the battlefield operations. Based on deployment and unattended nature of the sensor devices, make the underlying network vulnerable toward different attacks. Based on ad-hoc inherited nature and limited resources capabilities of wireless sensor devices, designing a strong and right keying scheme is very difficult. Most of recent approaches in literature existing considers trusted third parties as Certification Authority (CA). This is no longer applicable for the field of WSNs, because of lacking of knowledge about topology before deploying time of sensor devices. The current study is motivated by secure schemes proposed by (Zia and Zomaya

2006; Luk et al. 2007). Recently, researchers introduce great efforts in the field of key management schemes as in (Du et al. 2004; Du et al. 2006; Eschenauer and Gligor 2002; Chan et al. 2003). It has been proven that implementing a single keying algorithm is no longer feasible to provide multi-level services for all types of communication required by WSNs (Du et al. 2007).

Therefore, the promising keying mechanism includes different types of secret keys techniques, such as pre-deployed, within-network creation, and broadcast keys (Pradesh and Amberker 2009; Kocher et al. 2013). To overcome the existing shortcomings in literature, this work uses all mentioned keys types to provide adaptive security requirements. Furthermore, the hierarchically clustered based topology has been considered to manage the entire network communications as shown in Figure 1. Security measure does not

come free, the approach of adding heavy security measures poses substantial challenges in designing a light weight security solution proof attacks on WSNs.

In this paper we have considered three existing security techniques TinySec, MiniSec, and SHARP in terms of packet overheads and compared the time to transmit, average latency, and consumed energy with NSKM module. The comparison shows that the packet overheads in NSKM are fewer compared to other schemes. Our contributions in this paper are presenting NSKM module, an exhaustive analysis and evaluating of

the all aforementioned security techniques, and reinforced by the experimental results using BAN wireless sensor networks with Castalia-3.2 simulator.

The rest of the current work is oriented as: Section 2 presents most of existing related works in literature. In section 3, we discuss the functionality of the proposed NSKM module in details. Section 4 presents the comparative analysis of TinySec, MiniSec, SHARP, and NSKM schemes with experimental simulation results in section 5. Finally, the conclusions and some future works are introduced in section 6.

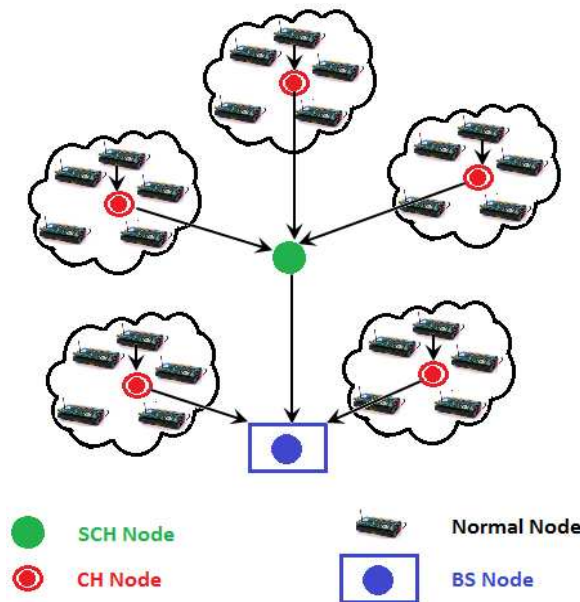


Fig. (1): Hierarchical cluster formation topology for WSNs

2. RELATED WORKS BACKGROUND

Recent approaches have been witnessed that complex cryptography techniques burden limited resources sensor devices (that is needing extra memory usage). In (Boyle and Newe 2008), several designing efforts for reaching optimal security model over WSNs have been debated in details. This paper infers that the symmetric cryptography outperforms an asymmetric from overheads perspective.

The work of (Perrig et al. 2002) achieves most of the security requirements, such as, two parties authenticity, confidentiality, authentication and integrity for broadcasting data packets.

The work of (Zhu et al. 2003) suggests a set of four secure keys techniques, to meet different security requirements upon various messages types.

To handle small size messages, 8 bytes secure key length is implemented in paper (Zhu et al. 2006). Based on this context, this approach provides robust security scheme for broadcasting packets within the entire network.

The work of (Zia and Zomaya 2006) implements a secure key management model to provide secure environment for WSNs. In general, this work is considered as motivation for the current research.

Several famous schemes, such as MiniSec, TinySec (Karlof et al. 2004), and SHARP(Sharma

et al. 2009) have been considered for comparison analysis of NSKM platform.

The approach in (Raman and Chebrolu 2008) suggested that functionality of WSNs is deeply based on underlying application scenarios. In this context, most of existing routing protocols are no longer handling any limited underlying application within its work for meeting the suggested cross-layer approach.

This work (Khalil et al. 2010) proposed cross-layer based solution for limited resource allocation for wireless sensor networks; it is no longer handled for security solution issues.

Authors of (Soroush et al. 2007; Sharma and Ghose 2009; Sharma and Ghose 2011) implement robust defense technique to node compromised attacks. To the best of our knowledge, the shortcoming presented within (Soroush et al. 2007) is no longer suitable for dynamic keying technique which needs periodically changing secret keys. Based on this, this work creates three keys types from the predefined key. Besides, there is no longer any efficient technique for isolating compromised sensor device within WSN.

With all mentioned drawbacks above in mind, the proposed NSKM exploits advantages presented in literature and removes some of their shortcomings like fixed keys method and communication overheads.

3. THE PROPOSED NSKM PLATFORM

Based on this study, depending on a single key technique is no longer suitable to secure all broadcasting types of messages used within WSNs. To overcome the existing shortcomings in literature, this work uses all mentioned keys types in section 1 to provide adaptive security requirements. The functionalities of the proposed platform include pre-creation, deployment, post-creation, and maintenance of the secret keys within entire network. In this context, A NSKM platform provides periodically renewing its secret keys to overcome likely compromised sensor devices. The secure hashing function SHA-1 is used to generate all post-deployed keys. The following subdivisions introduce the various types of secret keys used in this study and also discuss the master secret key deletion scheme.

A. Keys Types of the NSKM Platform

When the cluster members' discovery stage is done and the sensor nodes tables are updated, all

sensor nodes initiate the key creation stage. In general, we set up pair-wise key type technique among the sensor neighbors (Liu and Ning 2003). The technique considered in the proposed NSKM algorithm based on three types of keys: predefined key, in-network created keys, and BS broadcasted keys. Sets of secret keys used in the NSKM platform for securing the entire WSNs are:

- **Master key (K_M):** This key is predefined and preloaded into sensor devices within entire WSNs at offline. This secret key will be implemented for securing data within set up phase of the underlying secure routing protocol only. Once the set up phase is over, the K_M key must be deleted. That is for one hop (N:N) secure communication, where N is any node. The BS generates this key and preloads to all sensor nodes within the entire network.

- **My own key (K_O):** This type of key is used between any two sensor nodes initially. All the sensor nodes are preloaded with their Identifiers (ID), and the K_O is computed as in (1) which is the function of sensor node ID, BS ID, and master key K_M . The key K_O is implemented by all sensor nodes and destination node which is in our case BS (N: BS) for secure-communication. Also, it is implemented by a new sensor node adds to carry out network scalability. This type of key is in-network created key and all nodes generate this key by:

$$K_O = \text{SHA-1} \{ \text{MAX} (i \rightarrow \text{ID}, j \rightarrow \text{BS ID}) \parallel K_M \} \quad (1)$$

- **Cluster key (K_C):** This key is computed by all cluster member nodes within specific cluster. The K_C is implemented within every cluster (that is between cluster members and their CH node N: N). This type of key is important for all cluster members to follow up the cluster activities of neighbors. All the cluster members within specific cluster generate the key as in (2):

$$K_C = \text{SHA-1} \{ \text{MIN} (i \rightarrow \text{ID}, j \rightarrow \text{CH ID}) \parallel K_M \} \quad (2)$$

- **Super cluster key (K_{SC}):** This type of key is generated and issued by BS after nodes being authenticated as CHs and SCHs nodes., This set of secret key is only holding by SCHs and CHs sensor nodes only within current cycle of protocol process. The K_{SC} is implemented for SCH / CH: BS broadcasting messages. The BS node is responsible for refreshing this key at regular intervals of time and at the beginning of each round. The proposed NSKM platform is the functionality of application layer of the sensor node protocol

stack. Also this platform is considered as cross-layer based technique. (Gawdan et al. 2011).

B. Master Key K_M Deletion Scheme

After carrying out the initial secret keys establishment between all nodes within their clusters by implementing the proposed NSKM module, the master key K_M deletion phase is started. The deletion of K_M is done shortly after initial set up phase. Thus, there may be a small chance for opponent's participation in this time interim, and consequently there is no longer space for an opponent adding another sensor device to underlying network, because preloaded K_M has been totally removed regardless of the fact that there is a bargained node in the network.

C. Usage of the Secret Keys

The sensor devices are required to keep and follow up their secret keys. We assumed that the initial packet sent to the CH node by any node member is (M) and the packet is organized as:

$$\text{Node Message (NM)} = \{\text{ID}_{\text{SN}}, K_C, \text{TS}, \text{MAC}, (\text{M})\}$$

Where (TS) stands for time stamp implemented to prevent packets reiteration and (MAC) stands for Message Authentication Code. The first packet must be encrypted by K_C , to let all cluster members within specific cluster listen the sent packet for achieving the concept of cross-layer interactions. In this context, there arises a need to another secret key named K_{SC} . To get the K_{SC} , each of CH / SCH node broadcasts its encrypted request packet using its secret key K_O to BS. To this end, the BS broadcasts the encrypted K_{SC} -message by K_O including the determined K_{SC} to the requesting CH / SCH nodes. Therefore, after CH / SCH node collects its K_{SC} , all broadcastings through the SCH node take place using secret key K_{SC} and format of packet is organized as:

$$\text{CH Message (CM)} = \{\text{ID}_{\text{CH}}, K_{SC}, \{\text{ID}_{\text{SN}}, K_C, \text{TS}, \text{MAC}, (\text{NM})\}\}$$

Where, ID_{CH} stands for CH node identifier. To this end, all broadcastings to destination node BS take place via relaying and/or encrypting collected packets by super cluster key K_{SC} and the packet format is organized as:

$$\text{SCH Message (SCM)} = \{\text{ID}_{\text{SCH}}, K_{SC}, \{\text{ID}_{\text{CH}}, K_{SC}, \{\text{ID}_{\text{SN}}, K_C, \text{TS}, \text{MAC}, (\text{CM})\}\}\}$$

Where, the SCH node identifier is ID_{SCH} . To this end, the SCH node at once connects with the BS and adding its own ID_{SCH} to the collected packet from the sending CH node.

4. ANALYTICAL EVALUATION OF NSKM SCHEME

In general, the added overheads by the proposed NSKM scheme are analytically evaluated in the following subsections. There is trade-off between level of security and significant amount of the energy consumption in the field of WSNs. Unavoidable requirements of implementing security methods, like hashing process, process of encryption, process of decryption and keying technique are considered as consuming energy processes. In this context, the NSKM scheme also adds additional overhead which is storage space for saving the sets of secret keys.

A. Format of Packets

The format of packets used in existing approaches in literature and our proposed NSKM scheme is shown in Tables (1- 6). We carried out the performance comparison with, MiniSec, TinySec, and SHARP. All the values are measured in bytes. Based on Table 7 and Fig (2, 3 and 4), it obvious that the proposed NSKM module adds only 11bytes as overhead, which is outperforms others. The size of each packet is considered 40 bytes for the proposed NSKM module; such packet size can be implemented particularly with sensor devices currently provided in the market.

B. Computational Overhead

The comparison of security overheads added to the transmitted data for TinySec-Auth, TinySec-AE, SHARP, MiniSec, and NSKM schemes is depicted in Fig (2, 3 and 4) and Table 7. In this work, we consider data rate of 250-kbps for current calculations. Castalia simulator adds 35-bytes per sensor node as an overhead to the original size of each packet. The level of transmitted power (Tx) is chosen as 0-dBm (that is, the level of power consumption is selected as 57.42-mW (Boulis 2011)). Based on this comparison, it is clear that the proposed secret NSKM scheme outperforms the others in literature. Also does not have discernible additional overheads. TinySec considers the length of (≥ 8 -bytes) per message, in this context all messages (< 8 -bytes) would be neglected. Actually, Table 7 is an expanded version of our previous work (Gawdan et al. 2011) with more additional parameters and in-deep computational processes.

Table (1): Tiny OS: +7 bytes

Dest	AM	Group	Len	Data	CRC
(2)	(1)	(1)	(1)	(29)	(2)

Table (2): TinySec-Auth (Authentication only): +8 bytes

Dest	AM	Len	Data	MAC
(2)	(1)	(1)	(29)	(4)

Table (3): TinySec-AE (Authentication and Encryption): +12 bytes

Dest	AM	Src	Ctr	Len	Data	Mac
(2)	(1)	(2)	(2)	(1)	(29)	(4)

Table (4): MiniSec: +15 bytes

Dest	AM	PCF	DSN	Dst	Src	Len	Data	MAC(4)
(2)	(1)	(2)	(1)	PAN(2)	(2)	(1)	(29)	

Table (5): SHARP: +15 bytes

Dest	Keys	Track-id	Sect-id	Src	TS	Len	Data	MAC
(1)	(5)	(1)	(1)	(1)	(1)	(1)	(29)	(4)

Table (6): NSKM: +11 bytes

Dest	AM	Src	TS	Len	Data	MAC
(2)	(1)	(2)	(1)	(1)	(29)	(4)

Table (7): Overheads comparison among MiniSec, TinySec, SHARP, and NSKM Modules

	Application Data (bytes)	Packet Overhead (byte)	Total Size (bytes)	Time to transmit (ms)	Increase over TinyOS Stack (%)	Latency Overhead (%)	Energy Overhead (%)
Tiny OS	29	7	36	2.215	-	-	-
TinySec-Auth	29	8	37	2.246	1.4	1.39	1.38
TinySec-AE	29	12	41	2.371	7.04	7.04	7
MiniSec	29	15	44	2.464	11.26	11.24	11.25
SHARP	29	15	44	2.464	11.26	11.24	11.25
NSKM	29	11	40	2.34	5.6	5.6	5.6

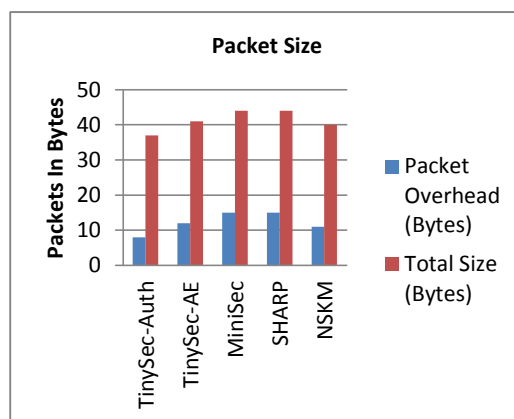


Fig. (2): Packet overheads comparison among TinySec-Auth, TinySec-AE, MiniSec, SHARP and proposed NSKM modules.

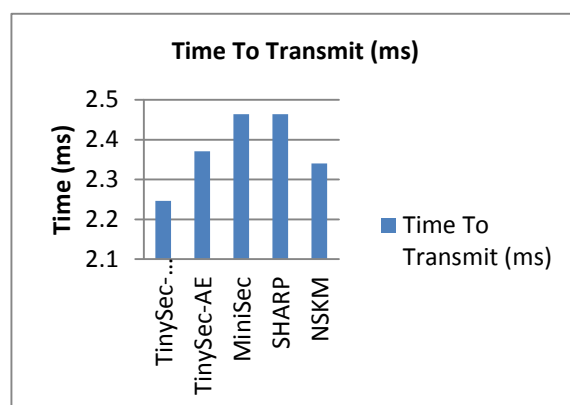


Fig. (3): Time to transmit packet overhead comparison among TinySec-Auth, TinySec-AE, MiniSec, SHARP and NSKM.

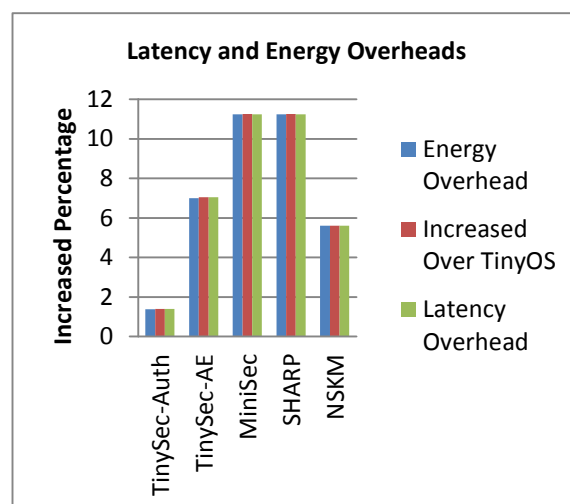


Fig. (4): Energy expenditure and latency comparison among TinySec-Auth, TinySec-AE, SHARP, MiniSec, and NSKM modules

C. Memory Storage Overheads

All cluster member devices hold only two sets of secret keys K_O and K_C . But when the sensor node is elected as CH node, it holds K_{SC} in addition to the above two sets of secret keys. While the SCH node holds the secret key K_O and various unique keys K_{SC} for different CH devices. This is applicable only for high level security required by underlying scenarios. In general, when the case of underlying application is no longer needing complex security scenarios; the task of SCH node is only to relay the collected packets toward the destination node which is the BS in our case. Thus, for most applications no longer need high level of security, the set of secret key K_{SC} is shared among all CHs and the SCH node for saving more energy. Based on our findings, for high level security scenario there is a need to increase the memory storage overhead. The storage space for normal node is expressed as:

$$\text{Node Storage} = [\text{Size of } (K_O) + \text{Size of } (K_C)] \quad (3)$$

The storage space for CH is:

$$\text{CH Storage} = [\text{Node Storage} + \text{Size of } (K_{SC})] \quad (4)$$

The storage space for SCH is as follow:

$$\text{SCH Storage} = [\text{Size of } (K_O) + \text{Size of } (K_{SC}) * X] \quad (5)$$

The variable X represents the elected CH sensor nodes number within entire WSNs. For instance, when length of underlying secret key is 10-bytes, the memory storage space for the cluster member nodes must be 20-bytes. It is not more than 30-bytes for a CH node, which is in general very small. Whilst the SCH in normal case that the underlying application requires low level of security, at most needs 20-bytes, due to using the shared key K_{SC} among all CHs with SCH. But for the unique secret key K_{SC} shared between SCH node and each CH node, the required storage of memory will be 510-bytes for (assume $X = 50$) CH sensor nodes within WSN. In this context, the above calculated memory storage space is applicable for available WSN nowadays.

5. SIMULATION RESULTS

We have implemented simulation in BAN scenario upon the wireless sensor network simulator named Castalia-3.2 (Castalia Website). The simulated scenario is done with 5 nodes,

varying parameter packet size (36, 40, and 44 bytes), and all nodes varying packet rate. Also using IEEE 802.15.4 MAC with its Guaranteed Time Slot (GTS) functionality is switched on. The simulation outcome is considered to compute the average number of the received packets for each node, latency level, and energy expenditure for different packet size of the Tiny OS, NSKM, and MiniSec / or SHARP, respectively. The packet size for both SHARP and Mini Sec has the same size 44 bytes. In this simulation, we consider 3 slots for every node (that is, the radio will be switched on only along these slots time; otherwise it will be switched off).

Functionality of GTS is a TDMA-based mechanism implemented by the standard protocol 802.15.4. For underlying application scenario, we consider for each round 16 slots time and the simulation time lasts for 50 seconds.

Figure 5 shows that the packets received are increasing with the increase of the packet rate for various schemes. However, our NSKM module out-performs the others for high traffic (data rate more than 26).

In Figure 6, most of the packets are received in the destination node during the first 120 ms which is considered as the MAC frame. Also one can observe a portion of packets which is very small with little delay in [600-inf] bucket; this is due to the potential saturation with overflow buffers. In addition, from observing the first [0-20] ms bucket, there are about (1050) packets having latency within [0-20] ms for NSKM module compared to (1015) packets within the same [0-20] ms latency for SHARP and MiniSec schemes. Consequently, it is obvious that NSKM module provides less latency with better performance compared to Mini Sec and SHARP.

From observing the consumed energy as depicted in Figure 7, it seems that the consumed energy no longer changes for the various schemes; since energy consumption depends on how long the radio is on. If the packet size does not affect this (and in many cases it does not), then there is no change in energy consumption. In GTS, which is TDMA-based, the nodes get a specific number of slots (in our scenario are 3 slots) that they can send and receive. Their radio is on during their slots and it is sufficient to send the different packets.

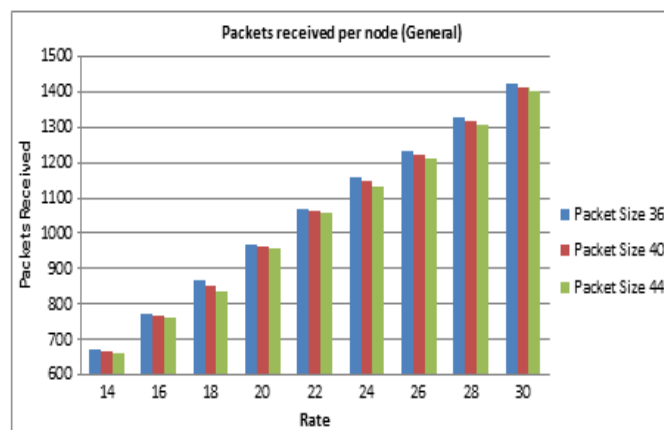


Fig. (5): Packets received per node by the sink 0 node

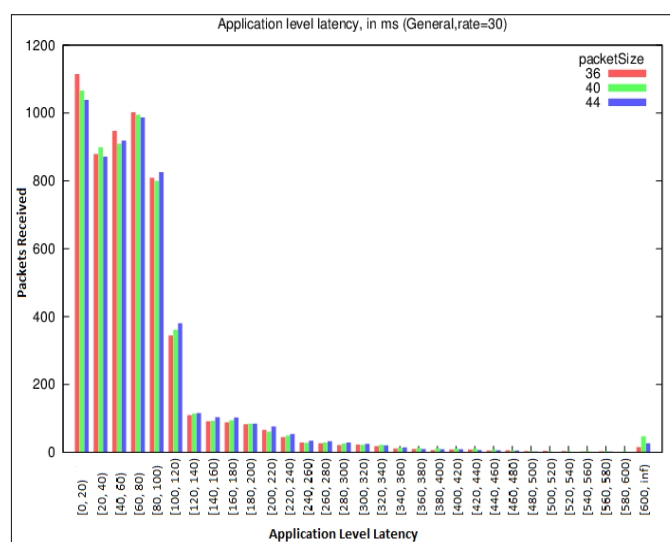


Fig. (6): Packets received against application level latency (buckets)

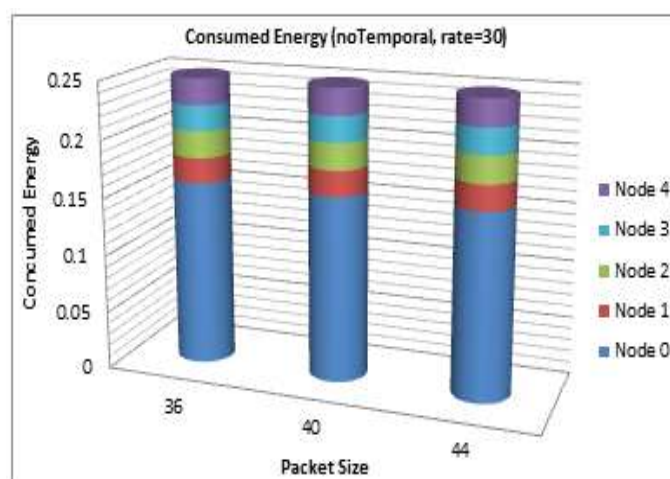


Fig. (7): Energy consumption (joules) for different size of packet.

6. CONCLUSION AND SOME FUTURE WORKS

The performance evaluation of NSKM platform is carried out in this study, which is considered as robust, an energy-aware, flexible, and lightweight keying technique that provides multi-level security services against likelihoods of adversary and presented attacks for WSNs. Based on the limited resources, hierarchically clustered formation approach and periodically maintaining keying technique, the entire WSN is never compromised even in the case of attendance of an attack in the network. In this work, an effort has been made to simulate and analyze NSKM scheme. The selection of SCH node based on both the remaining energy and its distance to the destination BS node, creation of in-network computed keys technique and using most communication types for wireless sensor node help the proposed NSKM module to hold and implement unique features (that is, cross-layer based approach).

From analytical comparison and experimental results, it is obvious that the proposed NSKM scheme provides fewer packet overheads, latency interval, and consumed energy, and it outperforms the existing approaches in literature. Besides, it adds very small increase over Tiny OS stack not more than 5.6 %, consequently.

We endeavor to use the proposed module in real-time and popular IRIS sensor nodes. There is a space for enhancing this work too:

- The CH \ SCH nodes need to compute their set of secret key K_{SC} within their own level instead of relying on the BS. In this context, there would be saving energy within the entire network.
- The proposed NSKM module will be considered as one of the essential building blocks of incoming future work.

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